Predicting Adolescents' Future Smoking Behavior through Anti-Smoking Ad Exposure: A Structural Equation Modeling Approach Based on the Theory of Planned Behavior.

By

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Abstract

Predicting Adolescents' Future Smoking Behavior through Anti-Smoking Ad Exposure: A Structural Equation Modeling Approach Based on the Theory of Planned Behavior.

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Introduction: The prevalence of cigarette smoking among adolescents in the U.S. is the lowest in any other past years, yet more than 8.1% still reported using tobacco products. Despite a downward trend in recent years, approximately 472,000 middle school and 729,000 high school students reported current cigarette use, highlighting the need for ongoing prevention initiatives. Utilizing the Theory of Planned Behavior, this study evaluates the impact of exposure to anti-smoking advertisements on adolescents' psycho-cognitive factors—specifically attitudes, social norms, perceived control, intentions, and smoking behavior. The findings aim to inform individuals about more effective and accessible public health interventions as part of the "smoke-free future" initiative.

Methods: A cross-sectional analysis was conducted using 2023 *Monitoring the Future* (MTF) survey data, a nationally representative sample of 8th and 10th graders. Structural Equation Modeling (SEM) was used to test relationships derived from the Theory of Planned Behavior, examining how anti-smoking ad exposure and psycho-cognitive factors predict current smoking behavior and intentions while adjusting for demographic variables.

Results: Ad exposure showed minimal direct effects on psycho-cognitive predictors of smoking. Instead, adolescents' risk perceptions toward tobacco use, perceived peer disapproval, prevalence of peers' tobacco use, and confidence in resisting smoking were stronger predictors of both smoking behavior in the past 30 days and intentions to not smoke in 5 years. Demographic factors, including parental presence, race, gender, and grade, significantly influenced adolescents' smoking-related perceptions and behaviors.

Conclusion: Psycho-cognitive factors are more influential than anti-smoking ads in predicting adolescents' smoking behavior. The findings emphasized the significance of involving peers and family influences in prevention strategies. SEM highlights the complex behavioral pathways involved, offering evidence-based recommendations for strengthening early, family-centered interventions to promote a smoke-free future generation.

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I want to extend my sincere gratitude to Dr. Weaver, Dr. Kirpich, and Boshi Wang for their invaluable support in the development of the thesis committee. I am especially thankful to Dr. Weaver for generously dedicating his time and sharing his expertise to design this comprehensive project. His guidance significantly deepened my understanding of the intricacies of Structural Equation Modeling. Boshi Wang has been exceptionally helpful in providing insightful guidance and thoughtful feedback to help me curate the most robust model possible. I also want to thank Dr. Kirpich for sparking my interest in statistical computing early in my academic career and encouraging the switch to Biostatistics concentration. Finally, I would like to thank my husband, parents, and in-laws for their unwavering support, and to my four-year-old son, whose cheerful presence brightened even the most challenging days.

Author's Statement

This thesis is submitted in partial fulfillment of the requirements for the Master's degree in Public Health, with a concentration in Biostatistics, at Georgia State University. I affirm that the work presented is entirely my own and was conducted with integrity, under the guidance and support of my thesis committee.

This research symbolizes the final step of my academic training in Public Health and Biostatistics. It provided a meaningful opportunity to apply advanced statistical tools and concepts to address real-world health challenges. The thesis focuses on adolescent substance use, specifically cigarette smoking as a critical public health issue due to its long-term health consequences and widespread impact. By studying smoking behavior among adolescents, this research aims to contribute to early prevention strategies and raise awareness about the dangers of tobacco use in youth populations.

The research highlights my proficiency to create advanced statistical models, manage large and complex datasets, ensure reproducibility, and outlining detailed, interpretable inferences. The research also emphasizes the competencies in statistical theory, research design, and applied data analysis within behavioral and population health frameworks. These skills have strengthened my capacity for evidence-based, data-driven decision-making and provided a solid foundation in modern Biostatistics developed throughout my graduate training.

Myesha Choudhury. I

Signed by Myesha Choudhury

Competencies Addressed in the Thesis

1. **Core Competencies**:

1.1. MPH 3. Analyze quantitative and qualitative data using biostatistics, informatics, computer-based programming, and software, as appropriate.

1.2. MPH 4. Interpret results of data analysis for public health research, policy, or practice.

1.3. MPH 22. Apply systems thinking tools to a public health issue.

2. **Concentration Competencies**:

2.1. MPH BSTP 3. Apply advanced (multivariate) descriptive and inferential techniques used with public health data.

2.2. MPH BSTP 8. Interpret results of statistical analyses found in public health studies.

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1. **Introduction**

Cigarette smoking among adolescents in the United States has declined to a historic low, with 2024 data showing the lowest prevalence since the National Youth Tobacco Survey (NYTS) began in 1999. [1] Nonetheless, tobacco use remains a persistent public health challenge. Over 2.22 million middle and high school students reported using tobacco products in the past 30 days, which included 5.4% of middle schoolers and 10.1% of high schoolers.[2] These figures underscore the need for sustained and targeted prevention efforts, particularly during adolescence, a critical period for brain development and maturation and a risk period for tobacco/nicotine experimentation, initiation, and addiction.[3] [4] One major contributor to youth's tobacco uptake has been the tobacco industry's pervasive marketing strategies. In 2019, U.S. cigarette advertising and promotional expenditures totaled $7.62 billion.[5] Widespread media portrayals reinforce the perception that smoking is associated with social approval and popularity.[6] Media representations in video games, movies, and celebrity coverage frequently depict smoking as socially acceptable or even glamorous, which can contribute to the normalization of tobacco use among young people.[7] [8] These portrayals may influence adolescents' perceptions of their peers' behavior and approval, potentially reinforcing both descriptive norms (the perceived prevalence of smoking among peers) and injunctive norms (the perceived disapproval of smoking by peers). [9]Anti-smoking messages can counter pro-smoking advertising and media portrayals to shift norms and shape attitudes and beliefs that affect intentions and behaviors regarding smoking initiation or cessation.[10]

This thesis is guided by the Theory of Planned Behavior (TPB), developed by Icek Ajzen, which provides a well-established framework for understanding human behavior by emphasizing behavioral intention as the most immediate predictor of behavior.[11] Behavioral intention is shaped by three central constructs: attitudes (positive or negative evaluation of the behavior), subjective norms (perceived social expectations from others to engage or not engage in the behavior), and perceived behavioral control (confidence in one's ability to perform the behavior). Subjective norms can be further categorized into injunctive norms, reflecting perceived peer approval or disapproval, and descriptive norms, reflecting the perceived prevalence of the behavior.[12] In the context of adolescent smoking, these constructs map onto specific cognitive beliefs: perceived risk (attitudes) towards smoking, perceived prevalence of peer using tobacco and peer disapproval of smoking (subjective norms), and confidence in resisting smoking (perceived behavior control). These factors influence behavioral intentions to not smoke for the next 5 years and, subsequently, future smoking behavior.

While TPB does not explicitly include anti-tobacco media campaign exposure as a predictor, anti-smoking advertising may serve as a catalyst that activates changes across its core constructs.[13] Public health campaigns commonly employ strategies to increase risk awareness, correct misperceptions about peer smoking, and promote refusal skills, aligning with TPB's mechanisms.[14] [15] Empirical evidence supports the effectiveness of such messaging: graphic warnings and social disapproval-themed campaigns have been shown to elevate perceived risk, reduce perceived peer acceptance of smoking, and decrease adolescents' intentions to smoke.[16] Thus, anti-smoking advertisements represent a key external influence that may indirectly affect current smoking behavior by reshaping the psycho-cognitive factors within the theory.

The gap this research fills is the lack of empirical studies that use Structural Equation Modeling to quantitatively evaluate how exposure to anti-smoking advertisements influences Theory of Planned Behavior constructs and smoking behavior among adolescents. While prior research has shown that media campaigns can affect risk perceptions towards smoking, and intentions to not smoke in 5 years, few studies have systematically modeled the direct and indirect pathways using nationally representative data, especially among adolescents. This thesis makes a novel contribution by integrating SEM with TPB to investigate how anti-smoking advertising influences attitudes, subjective norms, perceived behavioral control, intentions, and actual behavior. Thereby providing data-driven insights for designing more targeted and effective youth tobacco prevention campaigns.

H_0: Exposure to anti-smoking advertisements has no significant direct or indirect effects on adolescents' intentions to smoke or engage in smoking behavior. It does not significantly influence attitudes, subjective norms, or perceived behavioral control related to smoking.

H_1: Exposure to anti-smoking advertisements has significant direct and indirect effects on adolescents' smoking behavior by influencing key psycho-cognitive factors, such as attitudes, subjective norms, and perceived behavioral control, which in turn affect intentions to smoke and engage in smoking behavior.

2. **Methods**

2.1 Study Design

The present study employs a cross-sectional design, utilizing data from the 2023 Monitoring the Future (MTF) survey, which focuses on 8th and 10th graders, with a sample of approximately 14,000 respondents. The MTF survey is a nationally representative, serial cross-sectional study conducted annually since 1991.[17] It monitors trends in substance use, attitudes, and related behaviors among adolescents, college students, and adult high school graduates in the United States, to inform public health policy and prevention efforts.

While the MTF includes 8th-, 10th-, and 12th-grade students each year, this study excludes 12th-grade data due to structural complexities in the dataset. Specifically, the 12th-grade survey is divided into multiple questionnaire forms, leading to substantial item-level missingness and inconsistent variable coverage across respondents. These limitations reduce the feasibility of conducting a unified structural equation model for that subgroup.

The MTF utilizes a multi-stage, complex sampling design to ensure national representativeness.[18] Because this thesis involves only secondary analysis of publicly available, de-identified data, it does not meet the U.S. federal definition of human subject research and does not require IRB review.

2.2 Measures

This study is grounded in the Theory of Planned Behavior (Figure 1), expanded to include media exposure as the focal exogenous influence. The hypothesized model (Figure 2) consists of three latent constructs, four observed variables, and four demographic controls which include gender, parental presence, race, and grade level. The survey items and ordinal scale by constructs are provided in Table 1. Sample weights were added to the variables in the study to create a nationally representative sample of US adolescents, specifically in the eighth and tenth grades.

2.2.1 Outcome Measure

The primary outcome variable is current smoking behavior or smoking status, defined as cigarette use within the past 30 days. This is measured using a single observed binary indicator based on the question: “Have you smoked cigarettes in the past 30 days?” Responses were coded as 0 = No and 1 = Yes.

2.2.2 Primary Exposure Variable

Reported exposure to anti-smoking advertisements is a dichotomous observed variable constructed by combining two survey items: "How often have you seen anti-smoking commercials on TV or heard them on the radio?" and "How often have you seen anti-smoking ads on billboards or in magazines and newspapers?" Both items initially measured the frequency of exposure in recent months. For this study, responses were recoded into a binary indicator (1 = Yes, 2 = No) to reflect whether the adolescents had any exposure to anti-smoking media across these platforms. The recoding of the variable was aligned with the research objective, which focused on the presence or absence of exposure, rather than its frequency, to examine its association with TPB constructs and smoking behavior.

2.2.3 Putative Mediators

Perceived risk to health of tobacco use was assessed with three items: (1) “How much do people risk harming themselves physically and in other ways when they smoke one or more packs of cigarettes per day?” (2) “How much do people risk harming themselves physically and in other ways when they use any smokeless tobacco regularly?” (3) “How much do people risk harming themselves physically and in other ways when they smoke one to five cigarettes per day?" Each item was assessed on a four-point scale: 1 = No Risk, 2 = Slight Risk, 3 = Moderate Risk, 4 = Great Risk.

Injunctive norms for cigarette smoking were operationalized using two items assessing respondents' perception of their close friend's disapproval of smoking behavior: (1) "How do you think your close friends feel (or would feel) about you smoking one or more packs of cigarettes per day?" (2) "How do you think your close friends feel (or would feel) about you smoking cigarettes occasionally?" Each item was assessed on a four-point scale: 1="Not Disapprove" 2="Disapprove" 3="Strongly Disapprove".

Descriptive norms for tobacco use were operationalized using two items assessing respondents' estimate of how many of their peer smoke. Each item has a unique scale: (1) "How many of your friends would you estimate smoke cigarettes?" (2) "How many of your friends would you estimate use smokeless tobacco?" Each item was assessed on a five-point scale: 1="None" 2="A Few" 3="Some" 4="Most" 5="All".

Perceived behavior control to refrain from my smoking cigarettes was operationalized using a single item: "If one of your closest friends offered you a cigarette, how confident are you that you could refuse it?" Each item was assessed on a four-point scale: 1 = "Not at all confident" 2 =" Slightly confident" "3 = Moderately confident" 4 = "Very confident".

Intention to not smoke cigarettes in the next 5 years was operationalized using a single item: "Do you think you will be smoking cigarettes five years from now?" with responses measured on a four-point Likert scale: 1 = I definitely will, 2 = I probably will, 3 = I probably will not, 4 = I definitely will not.

2.2.4 Sociodemographic covariates

Gender was categorized as male or female and dummy coded, with female as the reference group. Parental presence was constructed by combining responses from two items: “Does your mother live in the same household with you?” and “Does your father live in the same household with you?” Responses were recoded into a single variable as follows: 0 = neither parent present, 1 = father only, 2 = mother only, and 3 = both parents present. This recoding was conducted for ease analysis and to examine the overall household parental presence in relation to exposure and psycho-cognitive factors, rather than modeling maternal and paternal presence separately. Race/ethnicity was categorized according to the MTF public-use dataset: Black, non-Hispanic; Hispanic; and White, non-Hispanic. All other responses were coded as missing by MTF investigators for de-identification purposes. Race/ethnicity was dummy coded, with White, non-Hispanic as the reference group. Grade level was dummy coded as 0 for 8th grade and 1 for 10th grade.

2.3 Analytical Plan

Analyses were conducted using Mplus Version 8.9,[19] following a stepwise process that included data preparation, measurement validation, and structural model testing. Structural Equation Modeling (SEM) was used to examine hypothesized relationships among TPB constructs, anti-smoking ad exposure, and smoking behavior. Prior to estimating the full SEM, Confirmatory Factor Analysis (CFA) was conducted to assess the measurement model for latent variables. Each construct was measured with three observed indicators, resulting in just-identified individual CFA models; therefore, factorial validity could not be formally assessed at that level. However, simultaneous estimation of all factors in a combined measurement model allowed for limited evaluation of factorial validity using model fit indices and standardized loadings. All indicators were treated as ordinal categorical variables, and estimation was performed using the Weighted Least Squares Mean and Variance adjusted (WLSMV) estimator with delta parameterization. WLSMV is robust to non-normality and appropriate for categorical outcomes, relying on polychoric correlations and a diagonal weight matrix.[20]

Latent variables were identified by fixing the first indicator loading to 1.0. Composite reliability was estimated using standardized loadings and error variances, with values of 0.70 or higher considered acceptable. The final analytic sample consisted of 13443 adolescents with complete data; cases with missing values were excluded through listwise deletion. Although the Monitoring the Future (MTF) survey uses a multistage complex sampling design, therefore, findings should be interpreted as representative of the weighted sample and interpret to the broader U.S. adolescent population.

2.3.1 Confirmatory Factor Analysis (CFA)

CFA was conducted to assess the reliability and factorial validity of the latent constructs. Model fit was evaluated using conventional thresholds: Comparative Fit Index (CFI) and Tucker-Lewis Index (TLI) ≥ 0.95, Root Mean Square Error of Approximation (RMSEA) ≤ 0.06, and Standardized Root Mean Square Residual (SRMR) ≤ 0.08.[21] Standardized factor loadings (β), Composite Reliability (CR > 0.70), and Average Variance Extracted (AVE > 0.50) were used to assess internal consistency and validity.[22]

2.3.2 Structural Equation Modeling (SEM)

SEM was used to test the hypothesized pathways among psycho-cognitive factors and exposure to anti-smoking ads on intention to not smoke in 5 years and current smoking behavior. Demographic variables were included in the structural model as covariates. The standardized coefficient (β) revealed the strength of the relationship. The model assumed β of approximately

0.10 to be a small effect, around 0.30 to be moderate, and above 0.50 to be considerable. Confidence intervals were estimated and reported.[23]

2.4 Ethical Considerations

This study used publicly available which only contained de-identified data from the MTF survey. IRB review was not required, and the original data collection adhered to federal ethical standards, including informed consent.[18] Principles of scientific integrity and transparency were upheld throughout the study. AI tools were used solely for editorial support and were reviewed for originality and accuracy.

3. Results

3.1 Sample Characteristics and Descriptive Statistics

Table 2 presents descriptive statistics for the analytic sample comprising 13443 observations, out of 14,734 respondents from MTF dataset due to Mplus's application of handling missingness with pairwise deletion. Also, some demographic data points are missing due to non-response or skipped to preserve privacy for PUF dataset. In the MTF survey dataset, approximately 44% identified as male, 44% as female with 11% missing. The racial composition was 39% White, 22% Hispanic, and 12% Black. Most participants were in the 10th grade (58%), with the remaining 42% in the 8th. Regarding household composition, 70% of respondents reported living with both parents, 16% with their mother, 3% with their father, and 1% with neither parent.

Table 3 presents univariate descriptive statistics for key model variables. These item-level distributions were examined to provide context for the latent constructs specified in the structural equation modeling framework.

The statistics showed that only 2% identified as current smokers in the past 30 days. Regarding intentions to smoke in 5 years, 72% expressed the lowest intention, whereas about 1% reported the highest intention to smoke. Regarding confidence in resisting smoking, 73% of respondents indicated the highest confidence level, while 2% reported the lowest confidence to refrain from smoking. Approximately 80% of respondents reported recent exposure to anti-smoking ads, and 20% reported no exposure. Most youth perceived a great risk of harm from smoking 1 or more packs per day (72%), a moderate or great risk of harm from regularly using smokeless tobacco (48%), and a moderate or great risk of harm from smoking one to five cigarettes per day (44%). Most adolescents perceived strong peer disapproval of smoking cigarette packs daily (68%) and had a moderate or higher perception of strong peer disapproval of smoking cigarettes occasionally (49%). Moreover, most adolescents perceived none of their peers smoked cigarettes (78%) and none used smokeless tobacco (84%).

3.2 Model Fit Evaluation and Use of Modification Indices

Table 4 illustrates the global fit indices for the measurement model. Although the Chi-Square statistic ($\chi^2 = 1086.447$, df = 58) is statistically significant, it is likely due to the large sample size of over 13,000 respondents. This assumption is standard in large-scale SEM, and it does not necessarily indicate poor model fit. Therefore, model fit is better evaluated using alternative indices, which suggest a good to excellent fit. The RMSEA of 0.036 indicates a close model fit, and both the CFI (0.975) and TLI (0.952) exceed conventional thresholds for acceptable fit. The SRMR of 0.090 is slightly above the ideal cutoff of 0.08 but remains within an acceptable range for complex models.

Modification indices (MIs) provided by Mplus were not used to revise the structural model. While MIs can help identify localized areas of misfit and suggest potential model improvements, applying them without theoretical justification may lead to overfitting, especially in the context of missing data. Given that the hypothesized model demonstrated strong global fit based on established thresholds, no post hoc modifications were implemented.

3.3. Confirmatory Factor Analysis (CFA) Results

Table 5 presents the standardized factor loadings for the latent constructs. Loadings ranged from 0.840 to 0.966 for risk perceptions of using tobacco, 0.898 to 0.979 for peer disapproval of smoking, and 0.818 to 0.960 for prevalence of peers using tobacco. These standardized factor loadings indicated strong associations between each observed indicator and its corresponding latent factor. All loadings were statistically significant at $p < .05$, though statistical significance is expected in large samples and does not necessarily imply practical significance or strong validity.

Internal consistency estimates for the latent constructs are reported in Table 6 using composite reliability (CR) and average variance extracted (AVE). CR values exceeded the commonly accepted threshold of 0.70, suggesting adequate internal consistency among indicators for each factor. AVE values indicated that approximately 82% of the variance in the indicators was explained by the attitudes construct, 59% by injunctive norms, and 53% by descriptive norms. While these values support acceptable measurement properties, the limited number of indicators (2–3 per construct) and the use of CFA alone place constraints on the strength of conclusions regarding construct validity.

3.4. Standardized Path Coefficients from the Structural Model

The structural path coefficients from the SEM are shown in Table 7. Exposure to anti-smoking advertisements was positively associated with a greater perceived risk of harm from tobacco use ($\beta = 0.347$, $p < .05$), stronger perceived peer disapproval of smoking ($\beta = 0.384$, $p < .05$), and lower perceived peer smoking ($\beta = -0.145$, $p < .05$). A negative association between exposure to anti-smoking ads and intention to not smoke in 5 years ($\beta = –0.253$, $p < .05$) indicated that exposure was linked to stronger intention to refrain from smoking for the next 5 years.

Intention to not smoke in 5 years was significantly predicted by confidence in resisting smoking ($\beta = 0.817$, $p < .05$), risk perceptions of using tobacco ($\beta = 0.201$, $p < .05$), peer disapproval of smoking ($\beta = 0.432$, $p < .05$), and low prevalence of peers using tobacco ($\beta = -0.343$, $p < .05$). Intention to not smoke in 5 years, in turn, was a strong negative predictor of smoking status (β = –1.384, $p < .05$). The path from smoking status to confidence in resisting smoking showed a positive and significant association (β = 0.711, $p < .05$).

Tables 8 and 9 summarize the effects of sociodemographic control variables on the model constructs. Significant associations were found between parental presence, grade level, gender, and racial/ethnic identity, and the constructs. For example, parental presence was positively associated with exposure to anti-smoking ads (β = 0.09, p = 0.003). Grade level (10th vs. 8th) was associated with higher risk perceptions of using tobacco (β = 0.128, p = 0.00), lower peer disapproval of smoking (β = -0.047, p = 0.00), higher prevalence of peers using tobacco (β = 0.114, p = 0.00), greater intentions to not smoke for 5 years, and current smoking behavior (β = 0.231, p = 0.00). Males reported lower risk perceptions of using tobacco (β = –0.059, p = 0.00) and lower prevalence of peers using tobacco (β = –0.059, p = 0.006). For Black adolescents, significant associations included lower risk perceptions of using tobacco (β = -0.057, p = 0.000), lower prevalence of peers using tobacco (β = -0.109, p = 0.000), lower intention to not smoke in 5 years (β = -0.102, p = 0.021), and higher confidence in resisting smoking (β = 0.168, p = 0.00). Among Hispanic adolescents, significant paths included lower risk perceptions of using tobacco (β = –0.072, p = 0.000) and lower number of respondents who recently smoked (β = –0.194, p = 0.003).

Overall, the model accounted for 96.7% of the variance in intention to not smoke in 5 years and 83.7% of the variance in current smoking behavior, indicating strong predictive power of the proposed TPB-based model. However, the model explained only 3.5% of the variance in confidence in resisting smoking and 1.3% in exposure to anti-smoking ads, suggesting limited explanatory power for these variables, likely due to the influence of demographic controls. Table 10 has more information on the explained variances.

4. Discussion

This study contributes to the growing literature on adolescent smoking prevention by providing a practical application of Structural Equation Modeling to evaluate how anti-smoking advertisement exposure influences constructs of the Theory of Planned Behavior and ultimately smoking behavior. While prior studies have examined the role of media in shaping adolescent behavior, few have quantitatively assessed the mediation pathways through TPB constructs using nationally representative data[3,18,24,25]. The current research addresses this gap by modeling how exposure to anti-smoking advertisements directly influences attitudes, injunctive norms, descriptive norms, and behavioral intentions. These psycho-cognitive factors, together with perceived behavioral control, predict intention, which in turn predicts smoking behavior.

The results affirm the study hypotheses. Exposure to anti-smoking messages was significantly associated with more negative attitudes toward smoking, stronger perceptions of peer disapproval (injunctive norms), and lower perceived smoking prevalence among peers (descriptive norms). These findings are consistent with prior evidence suggesting that adolescents' perceptions of peer behavior and approval are among the most influential predictors of substance use behaviors[24-26]. In this study, these normative constructs were predictive of lower intention to smoke, demonstrating the mediating mechanisms through which advertisements may affect behavior.

Importantly, the model explained a substantial proportion of variance in adolescents' intention to smoke, which underscores the predictive value of TPB constructs in tobacco-related decision-making. This aligns with prior findings that attitudes, subjective norms, and perceived behavioral control are robust predictors of behavioral intention across health domains[27,28]. The study's focus on 8th and 10th-grade students is also noteworthy, as this developmental period is critical for identity formation and susceptibility to peer influence.[3,4,24]

By integrating demographic moderators such as gender, race/ethnicity, parental presence, and grade level, the study offers a more nuanced understanding of how cognitive and environmental factors interact to shape adolescent smoking intentions. Previous research has shown that these demographic characteristics may moderate receptivity to public health messages and influence social norm perceptions[18,29]

4.1. Interpretation of Findings

The SEM model based on the 2023 Monitoring the Future dataset confirmed multiple directional pathways. Exposure to anti-smoking advertisements had a modest but statistically significant direct effect on intention to smoke in five years ($\beta = –0.253$). It was positively associated with risk perception, perceived peer disapproval, and a lower prevalence of peer tobacco use. These findings are consistent with earlier studies showing that anti-smoking campaigns influence youth behavior through changes in perceived norms and risk awareness.[25,30]

A strong positive correlation was found between behavioral intention and perceived behavioral control ($\beta = 0.817$), which suggests that adolescents who intend not to smoke also report more substantial confidence in resisting peer pressure to smoke. This supports the foundational assumptions of TPB, which posits that behavioral intention and self-efficacy together predict action.[27,28]

However, the model also revealed an unexpected positive association between current smoking status and self-reported confidence to resist smoking ($\beta = 0.711$). This contradicts theoretical expectations that nonsmokers would exhibit stronger perceived control. One explanation may be cognitive dissonance: adolescent smokers might overreport self-efficacy to reduce psychological discomfort caused by the contradiction between their behavior and internal beliefs.[31] Alternatively, this may reflect optimistic bias, whereby adolescents underestimate their vulnerability to tobacco dependence.[32,33]

These findings reinforce the importance of targeting psychosocial variables, particularly adolescents' subjective norms and their perceived behavioral control when developing youth tobacco prevention programs.[24]

4.2. Limitations of the Study

Several limitations should be considered when interpreting these findings. First, the cross-sectional and correlational nature of the study precludes causal inferences. Although SEM enables the testing of theoretically informed pathways, the directionality of effects cannot be confirmed without longitudinal or experimental research.[34]

Second, the reliance on adolescent self-report data introduces the risk of bias, including social desirability effects and inaccurate recall. These issues are especially relevant for questions about personal behavior and perceived norms among peer.[35] Third, one of the indicators used to measure the latent construct of perceived risk referenced smokeless tobacco use rather than cigarette smoking. Although the indicator demonstrated statistical fit within the measurement model, this inconsistency could affect construct validity and interpretability.

Fourth, the model accounted for a relatively small proportion of variance in perceived behavioral control and exposure to anti-smoking advertisements, indicating that these factors are influenced by additional variables not included in the analysis. Finally, since the analysis was limited to 8th and 10th-grade students enrolled in school, the findings may not generalize to older adolescents or out-of-school youth, who may have different exposure levels or motivational factors.

4.3. Implications for Future Research

Despite these limitations, the study offers important insights for public health practice and future research. The findings support a cost-effective and scalable prevention model that utilizes publicly available data and media-based messaging to influence adolescent behavior. Given the significance of subjective norms in predicting smoking intentions, future campaigns should emphasize peer disapproval and accurate representations of peer behavior to challenge misperceptions about smoking prevalence.[24]

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Appendix A: Mathematical Formulas[36]

1. CR (Composite Reliability) $= \frac{(\sum \boldsymbol{\lambda}_i)^2}{(\sum \boldsymbol{\lambda}_i)^2 + \sum \boldsymbol{\theta}_i}$
2. AVE (Average Variance Extracted) $= \frac{\sum \boldsymbol{\lambda}_i{}^2}{\sum \boldsymbol{\lambda}_i{}^2 + \sum \boldsymbol{\theta}_i}$

Appendix B: Additional Statistical Tables and Figures

1. List of Tables:

Table 1: Survey Items and Ordinal Scale by Construct

Construct	Item Code	Survey Question	Response Scale
Anti-Smoking Ads	ads	Have you seen or read about anti-smoking ads recently? (TV, radio, billboards, newspapers, or magazines)	1 = "No", 2 = "Yes"
Behavior: Smoking	smk	Have you smoked cigarettes in the past 30 days?	0 = "No", 1 = "Yes"
Attitudes	att1	How much do people risk harming themselves physically and in other ways when they smoke one or more packs of cigarettes per day?	1 = "No Risk", 2 = "Slight Risk", 3 = "Moderate Risk", 4 = "Great Risk"
	att2	How much do people risk harming themselves physically and in other ways when they use any smokeless tobacco regularly?	
	att3	How much do people risk harming themselves physically and in other ways when they smoke one to five cigarettes per day?	
Injunctive Norms	inj1	How do you think your close friends would feel about smoking one or more packs of cigarettes per day?	1 = "Not Disapprove", 2 = "Disapprove", 3 = "Strongly Disapprove"
	inj2	How do you think your close friends would feel about smoking cigarettes occasionally?	
Descriptive Norms	des1	How many of your friends would you estimate smoke cigarettes?	1="None" 2="A Few" 3="Some" 4="Most" 5="All"
	des2	How many of your friends would you estimate use smokeless tobacco?	

Perceived Behavioral Control	pbc	If one of your closest friends offered you a cigarette, how confident are you that you could refuse it?	1 = "Not at all confident" 2 =" Slightly confident" "3 = Moderately confident" 4 = "Very confident"
Behavioral Intention	int	Do you think you will be smoking cigarettes five years from now?	1 = "I definitely will", 2 = "I probably will", 3 = "I probably will not", 4 = "I definitely will not"
Demographics (Control)	plh	Is either your father or mother living in the same household as you?	0 = "No", 1 = "Father only", 2 = "Mother only", 3 = "Both"
	gender	How do you identify yourself?	1 = "Male", 2 = "Female"
	race	What is your racial/ethnic background?	1 = "Black or African American", 2 = "White (Caucasian)", 3 = "Hispanic"
	grade	Which grade are you in?	0 = "Eighth", 1 = "Tenth"

Table 2: Descriptive Statistics of Demographics

Constructs	Scale	Weighted (%)	Weighted N	Sample Size
Parental presence	No Parents	1.38%	186	12127
	Father Only	3.44%	462	
	Mother Only	15.89%	2136	
	Both Parents	69.50%	9343	
Gender	Male	44.30%	5955	11930
	Female	44.45%	5975	
Race	Black, non-Hispanic	11.66%	1567	9753
	White, Non-Hispanic	38.59%	5188	
	Hispanic, any race	22.30%	2998	
Grade-Level	8th	42.35%	5693	13443
	10th	57.65%	7750	

Note: Values are presented as counts and column percentages. Sample data are weighted.

Table 3: Descriptive Statistics in the Model

Variable	Sample Size	Scale	Category	Count	Proportion (%)
Intention to Not Smoke in 5 years	4549	1	I Definitely Will	25	1%
		2	I Probably Will	176	4%

		3	I Probably Will Not	1064	23%
		4	I Definitely Will Not	3284	72%
Confidence in Resisting Smoking	2289	1	Not at all confident	33	2%
		2	Slightly confident	99	4%
		3	Moderately confident	486	21%
		4	Very confident	1671	73%
Anti-Smoking Ad Exposure	4316	1	No Ad Exposure	879	20%
		2	Ad Exposure	3437	80%
Smoking Status	13431	1	Not Smoking	13224	99%
		2	Current Smoker	208	2%
Risk Perception of smoking 1+ packs of cigarettes per day	12784	1	No Risk	1089	9%
		2	Slight Risk	470	4%
		3	Moderate Risk	1993	16%
		4	Great Risk	9232	72%
Risk Perception of using smokeless tobacco regularly	11894	1	No Risk	1198	10%
		2	Slight Risk	1212	10%
		3	Moderate Risk	3758	32%
		4	Great Risk	5726	48%
Risk Perception of smoking 1-5 cigarettes per day	4363	1	No Risk	552	13%
		2	Slight Risk	360	8%
		3	Moderate Risk	1552	36%
		4	Great Risk	1900	44%
Peer Disapproval of smoking 1+ packs of cigarettes per day	4198	1	Not Disapprove	375	9%
		2	Disapprove	965	23%
		3	Strongly Disapprove	2858	68%
Peer Disapproval of smoking cigarettes occasionally	4219	1	Not Disapprove	660	16%
		2	Disapprove	1476	35%
		3	Strongly Disapprove	2083	49%
Prevalence of peers smoking cigarettes	8332	1	None	6466	78%
		2	A Few	1408	17%
		3	Some	316	4%
		4	Most	102	1%
		5	All	39	1%
Prevalence of peers using smokeless cigarettes	8310	1	None	6982	84%
		2	A Few	896	11%
		3	Some	324	4%
		4	Most	76	1%
		5	All	32	0%

Note: All variables were treated as categorical or ordinal in the analysis. Sample data are weighted.

Table 4: Global Fit Indices for the Model

Fit Index	Value	Threshold
Model χ^2 (df, p-value)	1086.447 (58, 0.00)	-
CFI	0.975	≥ 0.95
TLI	0.952	≥ 0.95
RMSEA	0.036	≤ 0.05
RMSEA 90% CI	[0.034, 0.038]	≤ 0.08
SRMR	0.090	≤ 0.08

Abbreviations: Chi-square test, χ^2. Comparative Fit Index, CFI. Tucker-Lewis Index, TLI. Root Mean Square Error of Approximation, RMSEA. Standardized Root Mean Squared Residual, SRMR.

Table 5: Standardized Factor Loading for Latent Constructs.

Construct	Indicator	λ (Standardized Loading)	S.E.
Risk Perceptions	Smoking 1+ packs of cigarettes per day	0.973	0.006
	Using smokeless tobacco regularly	0.840	0.007
	Smoking 1-5 cigarettes per day	0.904	0.007
Peer Disapproval	Smoking 1+ packs of cigarettes per day	0.898	0.022
	Smoking cigarettes occasionally	0.979	0.024
Prevalence of Peers Using Tobacco	Estimation of peers smoking cigarettes	0.960	0.034
	Estimation of peers using smokeless cigarettes	0.818	0.031

Note: All values are standardized factor loadings from confirmatory factor analysis. The *p-values* indicate statistical significance of factor loadings. All items shown are statistically significant ($p < 0.05$).

Table 6: Internal consistency of Measurements/ Scales.

Construct	CR (Composite Reliability)	AVE (Average Variance Extracted)
Risk Perceptions of Using Tobacco	0.93	0.82
Peer Disapproval of Smoking	0.94	0.59
Prevalence Peers Using Tobacco	0.89	0.53

Note: Values indicate internal consistency estimates based on confirmatory factor analysis. Composite reliability ≥0.70 and AVE ≥0.50 are commonly considered acceptable thresholds.

Table 7: Standardized Path Analysis of the Constructs in the Model

Path	β	S.E.
Confidence in Resisting Smoking → Intention to Not Smoke for 5 Years	0.817	0.018
Risk Perceptions of Using Tobacco → Intention to Not Smoke for 5 Years	0.201	0.029
Peer Disapproval of Smoking → Intention to Not Smoke for 5 Years	0.432	0.035
Prevalence of Peers Using Tobacco → Intention to Not Smoke for 5 Years	-0.343	0.031
Intention to Not Smoke for 5 Years→ Smoking status	-1.384	0.102
Exposure to Anti-Smoking Ads→ Intention to Not Smoke for 5 Years	-0.253	0.053
Confidence in Resisting Smoking → Smoking status	0.738	0.102
Exposure to Anti-Smoking Ads → Risk Perceptions of Using Tobacco	0.347	0.030
Exposure to Anti-Smoking Ads → Peer Disapproval of Smoking	0.384	0.034
Exposure to Anti-Smoking Ads → Prevalence of Peers Using Tobacco	-0.145	0.039

Note: Standardized path coefficients (β), standard errors (S.E.) presented in the table. All paths shown are statistically significant ($p < .05$).

Table 8: Path Coefficients of Latent Constructs with Sociodemographic Variables

Control (Reference Group in Parentheses)	Risk Perceptions of Using Tobacco			Peer Disapproval of Smoking			Prevalence of Peers Using Tobacco		
	β	**S.E.**	**p**	**β**	**S.E.**	**p**	**β**	**S.E.**	**p**
Male (Female)	-0.059	0.017	0.000	-0.037	0.026	0.155	-0.059	0.021	0.006
Hispanic, any race (White, non-Hispanic)	-0.072	0.018	0.000	-0.034	0.029	0.237	-0.006	0.024	0.808

Black, non-Hispanic (White, non-Hispanic)	-0.057	0.016	0.000	0.039	0.026	0.136	-0.109	0.022	0.000
10th Grade (8th Grade)	0.128	0.017	0.000	-0.047	0.027	0.000	0.114	0.022	0.000
Parental Presence in Household (None)	0.031	0.016	0.059	-0.005	0.029	0.853	-0.029	0.021	0.180

Note: Standardized path coefficients (β), standard errors (S.E.), and *p* values for associations between control variables and latent constructs are reported. Significant paths are indicated by $p < .05$.

Table 9: Path Coefficients of Observed Constructs with Control Variables

Control (Reference Group in Parentheses)	Intention to Not Smoke in 5 years			Confidence in Resisting Smoking		
	β	**S.E.**	***p-v*alue**	**β**	**S.E.**	***p-v*alue**
Male (Female)	-0.009	0.035	0.809	0.012	0.037	0.753
Hispanic, any race (White, non-Hispanic)	-0.029	0.038	0.437	-0.030	0.039	0.449
Black, non-Hispanic (White, non-Hispanic)	-0.102	0.044	0.021	0.168	0.044	0.000
10th Grade (8th Grade)	0.110	0.037	0.003	-0.056	0.038	0.143
Parental Presence in Household (None)	0.010	0.036	0.783	0.028	0.037	0.451
	Exposure to Anti-Smoking Ads			Smoking status		
	β	**S.E.**	***p-v*alue**	**β**	**S.E.**	***p-v*alue**
Male (Female)	-0.006	0.029	0.836	-0.022	0.054	0.686
Hispanic, any race (White, non-Hispanic)	-0.050	0.032	0.116	-0.194	0.065	0.003
Black, non-Hispanic (White, non-Hispanic)	-0.050	0.029	0.084	-0.078	0.062	0.214
10th Grade (8th Grade)	-0.011	0.031	0.731	0.231	0.058	0.000
Parental Presence in Household (None)	0.090	0.031	0.003	-0.015	0.051	0.763

Note: Table reports standardized path coefficients (β), standard errors (S.E.), and *p* values for associations between observed control variables and selected constructs. Bolded *p* values indicate statistical significance ($p < .05$).

Table 10: Explained Variance (R^2) for Observed and Latent Variables

Observed Variables	**Estimate**	**S.E.**	***p-v*alue**
Intention to Not Smoke in 5 years	0.967	0.049	0.000
Confidence in Resisting Smoking	0.035	0.016	0.030
Peer Disapproval of Smoking 1+ packs of cigarettes per day	0.803	0.039	0.000

Peer Disapproval of smoking cigarettes occasionally	0.961	0.047	0.000
Risk Perception of smoking 1+ packs of cigarettes per day	0.947	0.013	0.000
Risk Perception of using smokeless tobacco regularly	0.704	0.012	0.000
Risk Perception of smoking 1-5 cigarettes per day	0.817	0.013	0.000
Smoking Status	0.837	0.090	0.000
Prevalence of peers smoking cigarettes	0.940	0.067	0.000
Prevalence of peers using smokeless cigarettes	0.656	0.050	0.000
Exposure to Anti-Smoking Ads	0.013	0.007	0.051
Latent Variables	**Estimate**	**S.E.**	***p*-value**
Risk Perceptions of Using Tobacco	0.156	0.021	0.000
Peer Disapproval of Smoking	0.154	0.026	0.000
Prevalence of Peers Using Tobacco	0.050	0.014	0.000

Note: Table reports explained variance (R^2) estimates, standard errors (S.E.), and two-tailed *p* values. Values are based on structural equation modeling results.

2. List of Figures:

Figure 1: Theory of Planned Behavior Model

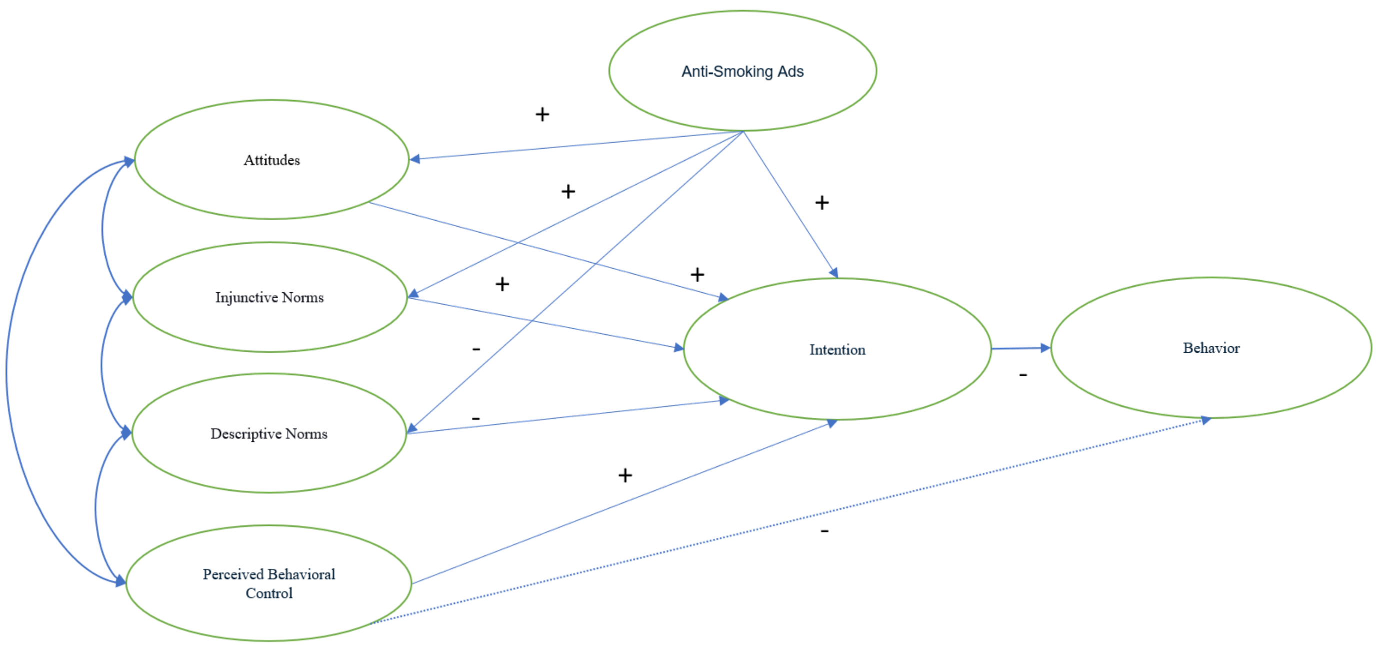

Figure 2: Hypothesized Model with Control Measurements